

AN ORTHOGONAL CLASS OF MODELS FOR TIME SERIES

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Summary: One of the most expensive aspects of fitting a model to time series data is that of model selection. Objective procedures are computationally expensive and subjective procedures require expertise. We propose a class of models and a discrepancy with respect to which the models are orthogonal. The discrepancy is based on the integrated squared difference between the spectral densities of the operating and the fitted model. The orthogonality property leads to a criterion for model selection which is simple and less expensive to compute. We show how the selected model can be used for forecasting and give a convenient estimator for the spectral density.

1. MODEL SELECTION AND ORTHOGONALITY

One of the most expensive aspects of fitting Autoregressive Moving Average (ARMA) models is that of model selection.

Objective selection procedures require considerable computing effort, so one relies mainly on subjective judgement - this requires expertise and time. The objective procedures (e.g. Akaike's (1969, 1973) AIC, FPE and Parzen's (1974, 1977) CATM) require that criteria be calculated for each competing model and the absence of orthogonality makes it necessary for one to fit each model individually. For example if one wants to select the order p of an AR model one has to re-estimate the parameters for each order p in order to compute the corresponding criterion. This is analogous to non-orthogonal polynomial regression, where one also has to re-estimate the regression coefficients if the order of the polynomial is changed.

The model families proposed here are *orthogonal*, and consequently the above difficulties do not arise. One can look at each parameter separately and decide whether it should be retained in the model. The expected discrepancy is simply the sum of components due to the individual parameters.

2. THE MODELS

The class of models which we consider are characterised by spectral densities of the form

$$g_{A,\theta}(\omega) = (1/2\pi)(\theta_0 + 2 \sum_{v \in A} \theta_v \cos v\omega),$$

where A is a set of p natural numbers. The parameters are restricted by the requirement that the matrix with elements $\{\theta_{|i-j|}\}$ is positive definite.

Each A determines a family of models which could be used for fitting and prediction. We consider the problems of model selection, forecasting and spectral estimation.

3. MODEL SELECTION

The model selection procedure used is based on the general framework outlined in Linhart and Zucchini (1982a, 1982b). Akaike's (1969, 1973) criteria FPE and AIC are well-known examples of criteria derived in this way.

The discrepancy between the operating model having spectral density $f(\omega)$ and the approximating family having density $g_{A,\theta}(\omega)$ which we use is based on

$$\int_{-\pi}^{\pi} (f(\omega) - g_{A,\theta}(\omega))^2 d\omega.$$

The functions $\{\cos \omega v\}$ are orthogonal with respect to this metric. Furthermore, the term $\int_{-\pi}^{\pi} f^2(\omega) d\omega$ obtained if the above expression is expanded does not depend on the approximating family. Consequently, for the purpose of model selection, this term may be omitted. We therefore use the discrepancy

$$\Delta(\theta) = -2 \int_{-\pi}^{\pi} f(\omega) g_{A,\theta}(\omega) d\omega + \int_{-\pi}^{\pi} g_{A,\theta}^2(\omega) d\omega.$$

We assume that the operating model is a Gaussian linear process with finite innovation variance, a spectral density which satisfies a Lipschitz-condition of order ξ and a covariance function $\gamma(v)$ with $\sum |\gamma(v)| < \infty$.

It is easy to show by means of well-known results that the parameter values, $\theta_0 = \{\theta_{0v}\}$, $v \in A^* = \{0\} \cup A$, which minimize this discrepancy are simply the covariances $\{\gamma(v)\}$, $v \in A^*$. The corresponding value of the discrepancy - the discrepancy due to approximation - is therefore

$$\Delta(\theta_0) = (-1/2\pi)(\gamma^2(0) + 2 \sum_{v \in A} \gamma^2(v)) .$$

As an empirical discrepancy we use

$$\Delta_T(\theta) = (2\pi/T) \sum_{j=0}^{T-1} (-2I_T(\omega_j) g_{A,\theta}(\omega_j) + g_{A,\theta}^2(\omega_j)) .$$

Here

$$I_T(\omega_j) = (1/2\pi) \sum_{v=-(T-1)}^{T-1} c(v) \cos \omega_j v$$

is the periodogramme, $\omega_j = 2\pi j/T$,

$$c(v) = (1/T) \sum_{t=0}^{T-1-v} x_{t+v} x_t$$

and x_t , $t = 0, 1, \dots, T-1$, (T odd) are the available data.

The minimum discrepancy estimator

$$\hat{\theta}_T = \arg \min_{\theta \in \Theta} \Delta_T(\theta)$$

is the vector with components $c(v)$, $v \in A^*$, and the minimal value of $\Delta_T(\theta)$ is

$$\Delta_T(\hat{\theta}_T) = (-1/2\pi)(c^2(0) + 2 \sum_{v \in A} c^2(v)) .$$

We calculate now the expected discrepancy, $E\Delta(\hat{\theta}_T)$. One has

$$T\Delta(\hat{\theta}_T) = T\Delta(\theta_0) + T(\hat{\theta}_T - \theta_0)' \Omega (\hat{\theta}_T - \theta_0)/2 ,$$

where

$$\Omega = \Delta''(\theta_0)$$

and (Anderson, 1971, Th. 8.4.2)

$$\lim_{T \rightarrow \infty} T^{\frac{1}{2}} E(\hat{\theta}_T - \theta_0)(\hat{\theta}_T - \theta_0)' = \Sigma,$$

where

$$\{\Sigma\}_{ij} = 4\pi \int_{-\pi}^{\pi} \cos i\omega \cos j\omega f^2(\omega) d\omega.$$

Hence

$$E\Delta(\hat{\theta}_T) = \Delta(\theta_0) + \text{tr } \Omega\Sigma/2T + O(1/T).$$

To obtain an estimator of this expectation we consider

$$T\Delta_T(\hat{\theta}_T) = T\Delta_T(\theta_0) - T(\hat{\theta}_T - \theta_0)'\Omega(\hat{\theta}_T - \theta_0)/2$$

$$E\Delta_T(\hat{\theta}_T) = E\Delta_T(\theta_0) - \text{tr } \Omega\Sigma/2T + O(1/T).$$

By Brillinger (1975, Th. 5.2.2) and Robinson (1978, p. 147)

$$E\Delta_T(\theta_0) = \Delta(\theta_0) + O(T^{-\min(1, \xi)})$$

and

$$\begin{aligned} E(\widehat{\text{tr } \Omega\Sigma}) &= E(-T/\pi) \sum_{v \in A^*} \widehat{\text{Var}} c(v) \\ &= E(-T/\pi) \sum_{v \in A^*} (4\pi^2/T) \sum_{j=0}^{T-1} \cos^2 \omega_j v I_T^2(\omega_j) \\ &= \text{tr } \Omega\Sigma + O(T^{-\min(1, \xi)}). \end{aligned}$$

An estimator of the expected discrepancy which is asymptotically unbiased to the order $O(T^{-\min(1, \xi)})$ is therefore

$$\begin{aligned} \hat{E}\Delta(\hat{\theta}_T) &= \Delta_T(\hat{\theta}_T) + (\widehat{\text{tr } \Omega\Sigma})/T \\ &= (-1/2\pi) \sum_{v \in A^*} c^2(v) + (1/\pi) \sum_{v \in A^*} \widehat{\text{Var}} c(v). \end{aligned}$$

According to this criterion a parameter, θ_v , should be retained in the model only if

$$c^2(v) > 2 \hat{\text{Var}} c(v) ,$$

where

$$\hat{\text{Var}} c(v) = (4\pi^2/T) \sum_{j=0}^{T-1} \cos^2 \omega_j v I_T^2(\omega_j) .$$

4. FORECASTING

The minimum mean square error forecast of x_T is $E(x_T | x_{T-1}, x_{T-2}, \dots) = \sum_i \phi_i x_{T-i}$ where the ϕ_i are the coefficients in the autoregressive representation of x_t . In practice one could approximate by setting $\phi_i = 0$ for $i > K$, where K is suitably large. The relationship between the ϕ_i and the parameters θ_v is given by the Yule Walker equations for an AR(K) process and estimators $\hat{\phi}_i$ are obtained by solving the system

$$\hat{\theta}_v = \sum_{i=1}^K \hat{\phi}_i \hat{\theta}_{v-i} , \quad v = 1, 2, \dots, K .$$

Note that $\hat{\theta}_v = 0$ for $v \notin A^*$. The forecast function is then

$$\hat{x}_{T-1}(1) = \sum_{i=1}^K \hat{\phi}_i x_{T-i} .$$

5. SPECTRAL ESTIMATION

The model selection procedure which results in the choice of a particular A (a particular approximating family) also provides a convenient estimator of the spectral density:

$$\hat{f}(\omega) = (1/2\pi)(c(0) + 2 \sum_{v \in A} c(v) \cos \omega v) .$$

6. DISCUSSION

The families of models considered here is best suited to time series of the "moving average type", i.e. which can be well approximated by parsimonious moving average representations. It is less suitable for series which are essentially autoregressive because such series usually do not have parsimonious moving average representations. Frequently one may circumvent this limitation by means of additional differencing of the series, for example the two representations AR(1) and ARIMA (1,0,1) often lead to very similar models.

We do not recommend that the procedure discussed here be uncritically applied to arbitrary time series. It should only be applied to situations where prior experience has shown that the given families do lead to parsimonious representations. The main feature of the procedure is that it provides a relatively cheap means of objective and automatic model selection. This is of advantage when one needs to fit many time series of a similar type, e.g. water or electricity consumption for many individual households, sales of several similar lines of consumer products, etc.

It is possible to apply the same discrepancy to a wider family of models or to use different discrepancies, (see Linhart and Volkers (1983), Volkers (1983) and the papers mentioned in section 1), but none of these enjoy the orthogonality property which allows for convenient model selection.

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